

Notes: Gaspar, Massa & Matos (JF, 2006)

Favoritism in Mutual Fund Families? Evidence on Strategic Cross-Fund Subsidization

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Sample and institutional setting	4
3.2	High-value and low-value fund classifications	4
3.3	Net-of-style returns	5
3.4	Actual pairs and matched pairs	5
3.5	Mechanism data: IPO allocations and opposite trades	5
4	Models and empirical specifications	5
4.1	Baseline same-family regression	5
4.2	Extended style-state regression	6
4.3	Family heterogeneity tests	6
4.4	IPO allocation tests	6
4.5	Opposite-trade regressions	7
5	Identification and empirical design	7
5.1	Core identification problem	7
5.2	Actual versus matched pairs	7
5.3	Controls and fixed effects	7
5.4	Mechanism tests as credibility checks	7
5.5	What the design cannot fully prove	8
6	Section-by-section study guide	8
7	Tables and figures	8
7.1	Table I: Characteristics of High and Low Funds	8
7.2	Table II: Univariate Analysis	9
7.3	Table III: Tests of Strategic Cross-Fund Subsidization	9
7.4	Table IV: Extended Tests of Strategic Cross-Fund Subsidization	10
7.5	Table V: Family Characteristics and Strategic Cross-Fund Subsidization	11
7.6	Table VI: IPO Allocations across High and Low Funds	12
7.7	Table VII: Opposite Trades and Strategic Cross-Fund Subsidization	12
8	Detailed result map	13

9 Short summary	13
10 Conclusion	14
10.1 Core conclusion	14
10.2 Economic intuition	14
10.3 Incremental contribution recap	14
10.4 Mechanism and evidence assessment	14
10.5 Regulatory implications	14
10.6 Investor implications	15
10.7 Limitations	15
10.8 Final takeaway	15

1 Big picture

- **Question.** Do mutual fund families strategically reallocate performance across their member funds, favoring funds that are more valuable to the management company even when this harms investors in other funds in the same family?
- **Core mechanism.** A mutual fund family maximizes family-level fee income and franchise value, not necessarily the risk-adjusted return of every individual fund. This creates incentives to help certain “high family value” funds at the expense of “low family value” funds.
- **High-value funds.** The paper studies three dimensions of high family value: high-fee funds, high year-to-date performers, and young funds. High-fee funds generate more direct fee revenue; high-performing funds attract convex investor inflows and create star-fund spillovers; young funds may have stronger flow-performance sensitivity.
- **Low-value funds.** Low-fee, low-performing, and old funds are less valuable at the margin to the management company, so they may become the source of implicit subsidies.
- **Main empirical strategy.** The paper constructs high/low fund pairs and compares the net-of-style performance gap inside the same family to the gap for matched pairs of similar funds from different families.
- **Main result.** High-value funds inside the same family outperform low-value funds by more than comparable high/low pairs from different families. The implied in-family favoritism is roughly 6–28 basis points per month depending on the criterion.
- **Mechanism evidence.** Better allocations of underpriced IPOs and opposite trades across member funds partially explain why high-value funds outperform.
- **Where favoritism is stronger.** Subsidization is more prevalent in large families, families with many funds, and families with heterogeneity in fund size, because these families have more opportunities and stronger incentives to reallocate performance internally.
- **Investor implication.** Investors in favored funds may benefit, but investors in disadvantaged funds bear hidden costs that are difficult to detect from fund-level performance alone.
- **Regulatory implication.** The results speak directly to debates over cross-trading and intra-family allocation practices, where cost-saving benefits must be weighed against self-dealing risk.
- **Economic takeaway.** Delegated asset management is not only a fund-level agency problem. It is also a family-level agency problem: the parent organization may distort returns across funds to maximize the value of the fund complex.

2 Incremental contribution of the paper

- **Moves beyond fund-level incentives.** Earlier mutual fund research mainly studies incentives of individual portfolio managers, such as risk-taking, tournament behavior, and flow-performance incentives. This paper shifts the unit of analysis to the *fund family*, showing that the family can coordinate behavior across funds.
- **Introduces strategic cross-fund subsidization.** The paper defines and tests a new form of agency distortion: performance can be transferred from one fund to another inside a family when doing so increases family value.
- **Connects mutual fund organization to internal capital markets.** The paper is analogous to internal capital-market work in corporate finance. Instead of reallocating capital across divisions, a mutual fund family reallocates performance opportunities, IPO allocations, and trading advantages across funds.

- **Develops an actual-pair versus matched-pair design.** The paper’s identification strategy compares high/low funds within the same family to high/low funds of similar style and rank from different families. This is an important methodological contribution because it tries to separate in-family favoritism from general differences between high- and low-value funds.
- **Documents both performance effects and mechanisms.** The paper does not stop at return gaps. It tests two concrete channels: preferential IPO allocation and opposite trades, which are plausible operational mechanisms through which families can shift performance.
- **Shows heterogeneity by family structure.** The paper links favoritism to family characteristics, showing that large, broad, and heterogeneous families are more likely to subsidize strategically. This helps explain why family organization matters.
- **Adds a regulatory contribution.** The paper provides evidence relevant to cross-trading rules and allocation policies. It shows that practices that may reduce transaction costs can also enable self-dealing across investors in different funds.
- **Adds a welfare contribution.** The paper clarifies that even if fund families generate economies of scale and scope, these benefits can coexist with hidden redistribution across investors. The family form can therefore be both value-enhancing and agency-distorting.

3 Data and measurement

3.1 Sample and institutional setting

- The sample consists of actively managed U.S. equity mutual funds belonging to the top 50 fund families in the CRSP Mutual Fund database.
- The sample period is 1991–2001.
- The paper focuses on actively managed equity funds because these funds give managers and families discretion over trades, allocations, and performance-affecting decisions.
- Index funds are excluded because their objective is mechanical tracking and there is less scope for discretionary performance shifting.
- The top families account for a very large fraction of mutual fund assets, making the family organization economically important.

Interpretation: The paper studies a setting where individual funds are legally separate investment vehicles but are operated by a common management company. This creates a natural environment for conflicts between fund-specific investors and the family-level objective.

3.2 High-value and low-value fund classifications

Each month, funds are partitioned into high and low groups along three criteria:

- **Total fees.** A fund is high value if its total fees are above the 75th percentile within its family and low value if below the 25th percentile. High-fee funds generate more fee income per dollar of assets.
- **Year-to-date return.** A fund is high value if its year-to-date return is above the 75th percentile within its style and low value if below the 25th percentile. High-performing funds are more likely to receive investor inflows.
- **Age.** A fund is high value if it is young and low value if it is old. Young funds are considered potentially valuable because the flow-performance relation may be stronger for newer funds.

Interpretation: These definitions translate the family’s objective into empirical rankings. The family has stronger incentives to improve the performance of funds with high direct fees, strong recent performance, or potentially high future flow sensitivity.

3.3 Net-of-style returns

For each fund, the paper computes a net-of-style return:

$$\text{Net-of-style return}_{i,t} = R_{i,t} - R_{\text{style}(i),t}.$$

The style benchmark is based on the fund’s ICDI investment objective code.

Interpretation: Net-of-style returns remove broad style movements. If growth stocks rise, all growth funds may benefit; the net-of-style return asks whether a particular fund outperformed other funds in the same style.

3.4 Actual pairs and matched pairs

- **Actual pairs:** high-value and low-value funds that belong to the same family.
- **Matched pairs:** the same high-value fund is paired with a control low-value fund from outside the family, matched by style and the decile of the relevant classification variable.
- The dependent variable is the difference in net-of-style returns between the high-value fund and the low-value fund.

Interpretation: The matched-pair design is central. If high-fee funds naturally outperform low-fee funds for reasons unrelated to family favoritism, the matched pairs should also show a similar performance gap. The family-specific component is the extra gap in actual pairs.

3.5 Mechanism data: IPO allocations and opposite trades

- IPO data come from SDC and are merged to mutual fund holdings.
- The paper measures IPO underpricing using first-day returns from offer price to first-day closing price.
- A fund is treated as receiving an IPO allocation if it holds the IPO shortly after issuance.
- Mutual fund holdings data from Spectrum/Thomson Financial are used to identify portfolio positions and trades.
- Opposite trades are measured when high- and low-value funds in the same family trade the same stock in opposite directions, either contemporaneously or with lagged timing.

Interpretation: IPO allocations and opposite trades are not merely auxiliary tests. They provide direct evidence on mechanisms through which the family can transfer value across funds.

4 Models and empirical specifications

4.1 Baseline same-family regression

The main regression is:

$$\text{NetReturn}_{i,t}^{\text{High}} - \text{NetReturn}_{j,t}^{\text{Low}} = \alpha + \beta \text{SameFamily}_{ij} + \gamma \text{SameStyle}_{ij} + \text{Controls}_{ij,t} + \varepsilon_{ij,t}.$$

- The dependent variable is the performance gap between a high-value and a low-value fund.
- SameFamily equals one for actual pairs and zero for matched pairs.
- β measures the extra performance gap inside the same fund family.
- A positive β is evidence consistent with strategic cross-fund subsidization.

Interpretation: The coefficient β is the key favoritism measure. It asks whether high-value funds outperform low-value funds more when both funds belong to the same family than when a similar pair is formed across families.

4.2 Extended style-state regression

The extended regression separates cases where the high-value fund's style is outperforming or underperforming the low-value fund's style:

$$\begin{aligned}\Delta NetReturn_{ij,t} = & \alpha + \beta_+ (\text{SameFamily}_{ij} \times 1\{STRET_H > STRET_L\}) \\ & + \beta_- (\text{SameFamily}_{ij} \times 1\{STRET_H < STRET_L\}) \\ & + \beta_0 (\text{SameFamily}_{ij} \times \text{SameStyle}_{ij}) + \gamma \text{SameStyle}_{ij} + Controls + \varepsilon_{ij,t}.\end{aligned}$$

Interpretation: This test asks whether families subsidize more when the low-value fund's style is doing well. If low-value funds have style-driven gains, there is more performance to transfer or offset.

4.3 Family heterogeneity tests

The paper repeats the baseline regression within family groups:

- top 25 versus bottom 25 families by size;
- above-average versus below-average number of funds;
- old versus young families;
- above-average versus below-average dispersion in fund size.

Interpretation: Larger and more complex families have more funds across which to reallocate performance and stronger incentives to manage family-level brand and flow externalities.

4.4 IPO allocation tests

The IPO tests compare whether high-value funds receive more underpriced IPOs than low-value funds within the same family. Outcomes include:

- number of IPOs held;
- average and median first-day IPO returns;
- dollar amount of underpricing allocated;
- percentage contribution of IPO underpricing to fund returns.

Interpretation: Underpriced IPOs are scarce positive-NPV opportunities. If families allocate more hot IPOs to high-value funds, the allocation mechanism directly shifts performance.

4.5 Opposite-trade regressions

The opposite-trade regression adds measures of trading in the same stock by high- and low-value funds in the same family:

$$\Delta NetReturn_{ij,t} = \alpha + \beta \text{SameFamily}_{ij} + \gamma \text{SameStyle}_{ij} + \theta \text{OppositeTrades}_{ij,t} + \text{Controls} + \varepsilon_{ij,t}.$$

Interpretation: Opposite trades can reflect coordinated liquidity provision or direct cross-trading. A negative or attenuating effect of opposite-trade controls on the same-family coefficient suggests that such trades explain part of favoritism.

5 Identification and empirical design

5.1 Core identification problem

The key empirical challenge is that high-value and low-value funds may differ for reasons unrelated to favoritism. High-fee funds, winning funds, and young funds may naturally differ in skill, risk, style, liquidity, or investor base.

Interpretation: The paper does not claim random assignment of funds to high- and low-value categories. Instead, it uses matched comparisons and fixed effects to isolate the family-specific component of performance differences.

5.2 Actual versus matched pairs

- Actual pairs compare high- and low-value funds in the same family.
- Matched pairs preserve the high-value fund but replace the low-value fund with a similar low-value fund from outside the family.
- Matching is based on style and the decile of the classification variable.

Interpretation: The matched-pair comparison is the central identification device. If the extra return gap only appears inside families, the evidence points toward family-level strategic behavior rather than general differences between high- and low-value funds.

5.3 Controls and fixed effects

The regressions include controls for fund size, fund age, family size, family age, year dummies, family dummies, and style dummies.

Interpretation: These controls reduce concerns that the same-family coefficient simply captures permanent family quality, style composition, or time-varying market conditions.

5.4 Mechanism tests as credibility checks

- IPO allocation tests ask whether scarce underpriced deals are directed toward high-value funds.
- Opposite-trade tests ask whether trades across funds inside the family contribute to the observed performance gaps.

- Family heterogeneity tests ask whether favoritism is stronger where incentives and opportunities are stronger.

Interpretation: A return-gap result alone could be hard to interpret. The mechanisms and heterogeneity patterns make the strategic-subsidization interpretation more credible.

5.5 What the design cannot fully prove

- The paper cannot observe the internal decision process of fund families.
- It cannot prove that every opposite trade is intentional self-dealing rather than independent portfolio choice.
- It cannot perfectly observe all execution quality, trade timing, and internal allocation rules.
- It cannot fully separate strategic favoritism from unobserved skill differences across portfolio managers.

Interpretation: The evidence is circumstantial but multifaceted: performance gaps, IPO allocations, trading patterns, and family heterogeneity point in the same direction.

6 Section-by-section study guide

- **Introduction.** Defines the family-level agency problem and motivates why families might favor high-fee or high-performing funds.
- **Hypotheses.** Derives incentives to subsidize high-value funds and proposes mechanisms such as IPO allocation and opposite trades.
- **Data.** Describes CRSP mutual funds, top 50 families, style classifications, total fees, age, returns, and Spectrum holdings.
- **Main tests.** Compares actual within-family high/low pairs to matched across-family pairs.
- **Extended tests.** Studies when favoritism is stronger, including style-state and family-characteristic splits.
- **Mechanisms.** Tests preferential IPO allocation and opposite trading across family funds.
- **Economic magnitude.** Assesses how much mechanisms can explain and how much fund families may gain from redistribution.
- **Conclusion.** Emphasizes family-level strategic behavior and regulatory implications for cross-trading.

7 Tables and figures

7.1 Table I: Characteristics of High and Low Funds

Read:

- Table I reports summary characteristics for high- and low-value funds under the total fees, year-to-date return, and age classifications.
- High-fee funds have total fees of 2.2% versus 0.9% for low-fee funds.
- High year-to-date return funds have much higher year-to-date returns than low performers, as expected from the classification.

- Under the age criterion, high-value funds are young funds with average age 0.8 years versus 12 years for old funds.
- Fund size differs substantially in some classifications, so controls for fund size are important in later regressions.

Detailed interpretation: The table establishes that the high/low groups are economically meaningful. The high-fee classification captures funds that generate more revenue per dollar of assets. The year-to-date return classification captures funds that are positioned to attract performance-sensitive flows. The age classification tests whether young funds are favored, although later results show weaker support for this channel.

Economic message: The table maps the family-value concept into observable fund characteristics. Without this mapping, the favoritism hypothesis would be too abstract to test.

each line presents the sample-wide mean of each variable for high funds, the mean for low funds, and the p -value of the hypothesis test that the two means are equal.

	Total Fees			Year-to-Date Return			Age		
	High Funds	Low Funds	p -Val. Diff.	High Funds	Low Funds	p -Val. Diff.	High Funds	Low Funds	p -Val. Diff.
Fund return ^a	0.011	0.011	0.999	0.014	0.007	<0.001	0.010	0.010	0.943
TNA	641	2,105	<0.001	1,798	1,330	<0.001	1,042	3,603	<0.001
Total fees	0.022	0.009	<0.001	0.014	0.016	<0.001	0.015	0.013	<0.001
Year-to-date return ^a	0.008	0.010	<0.001	0.020	-0.003	<0.001	0.008	0.009	0.113
Age	2.1	4.7	<0.001	3.9	3.6	<0.001	0.8	12.0	<0.001

Table I: Characteristics of High and Low Funds

t -statistic of the test that the two means are equal. The symbols ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Net Difference in Performance for Pairs of High and Low Funds		
	Actual Pairs	Matched Pairs	p -Value of Difference
Total fees	0.052%***	0.000%	0.03**
Year-to-date return	0.665%***	0.492%***	<0.001***
Age	0.004%	-0.021%	0.25

Table II: Univariate Analysis

7.2 Table II: Univariate Analysis

Read:

- Table II compares actual within-family pairs with matched across-family pairs.
- For total fees, actual pairs show a 0.052% monthly performance gap, while matched pairs show essentially zero.
- For year-to-date return, actual pairs show a 0.665% gap, while matched pairs show 0.492%; the difference is statistically significant.
- For age, the actual and matched differences are small and not meaningfully different.

Detailed interpretation: The univariate results provide the first evidence of same-family favoritism. The total-fee result is especially clean because matched pairs show no comparable gap. The year-to-date return result shows that winners outperform losers generally, but the same-family setting adds an extra increment. Age is not supported in this simple comparison.

Economic message: The table shows that performance gaps are not simply a property of high-value funds. The gaps are larger inside families, consistent with family-level intervention.

7.3 Table III: Tests of Strategic Cross-Fund Subsidization

Read:

- Table III estimates the baseline regression of high-minus-low net-of-style returns on the Same Family indicator.

- For total fees, the same-family coefficient is positive and statistically significant once controls are included.
- For year-to-date return, the same-family effect is large and highly significant: high-performing funds receive extra performance relative to low performers inside the same family.
- For age, coefficients are negative or insignificant, providing little support for favoritism toward young funds.
- The estimates include year, family, and style dummies and have very large numbers of pair-month observations.

Detailed interpretation: The baseline multivariate tests are the core return evidence. The coefficient on Same Family measures the incremental high-low performance gap inside the same family after comparing against matched pairs and controlling for observable differences. The strongest evidence is for high-fee and high-performing funds. The age channel does not behave as predicted, suggesting that families do not generally sacrifice old funds for young funds.

Economic message: Families appear to subsidize funds with clear revenue or flow value. The evidence is less consistent with a blanket preference for launching or supporting young funds.

Table III
Tests of Strategic Cross-Fund Subsidization

This table presents regression results of the tests of cross-fund subsidization. Every month, we partition funds into quartiles with respect to a variable of interest (e.g., Total Fees). A fund is classified as High (Low) if the fund's value for that variable is above (below) the 75th (25th) percentile for the relevant peer group of funds. For Total Fees and Age, the comparison peer group is all funds in the same family. For Year-to-Date Return, the comparison peer group is all funds in the same style (ICDI investment objective code). Note that for the Age criteria, a High value fund is a young fund, while a Low value fund is an old fund. We then construct, for each classification, two sets of High/Low pairs of funds as follows. In the first set, each High fund is matched with all Low funds belonging to the same family. We call these pairs, constructed within the same family, "Actual" pairs. In the second set, each Low fund in every Actual pair is replaced by a matching control fund taken from the remaining sample of funds. The matching control fund is selected randomly from the set of funds of the same style and belonging to the same decile of the variable of interest as the Low fund it replaces (Total Fees, Year-to-Date Return, or Age). We call these pairs "Matched" pairs. For each fund in the pair, we calculate the Net-of-Style Return (defined as the fund's monthly return minus the return for its style) and subsequently the Difference in Net-of-Style Returns, the difference between the Net-of-Style Return of High fund i and Low fund j within each pair. Both sets of pairs are added together in the same data set to run our regressions. For each classification, the table shows the results of the following regression (equation (1) in the text):

$$\text{Net.return}_{i,t}^{\text{High}} - \text{Net.return}_{j,t}^{\text{Low}} = \alpha + \beta(\text{Same.family}) + \gamma(\text{Same.style}) + \text{controls} + \varepsilon_{i,j,t}.$$

The left-hand side variable is the Difference in Net-of-Style Returns for each pair. Same-Family is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same fund family, and 0 otherwise. Same-style is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same style, and 0 otherwise. Control variables whose coefficients are not shown include, for both the High and the Low funds in each pair, the size of the funds, the age of the funds, the size of the funds' families and the age of the funds' families. t -statistics shown are obtained using heteroskedastic-robust standard errors. The symbols ***, **, and * denote significance at 1%, 5%, and 10% levels, respectively.

	Total Fees				Year-to-Date Return				Age			
	(1)		(2)		(3)		(4)		(5)		(6)	
	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.
Intercept	0.0012	2.01*	0.0019	2.78*	-0.0007	-0.65	-0.0006	-0.48	-0.0014	-1.81*	-0.0009	-0.93
Same family (β)	0.0004	1.75*	0.0006	2.44**	0.0021	5.68***	0.0028	6.49***	-0.0004	1.95*	-0.0003	-1.33
Same style (γ)	-0.0003	-1.06	-0.0003	-1.06	0.0011	2.42**	0.0011	2.46**	-0.0001	-0.20	-0.0001	-0.24
Controls	no		yes		no		yes		no		yes	
Year dummies	yes		yes		yes		yes		yes		yes	
Family dummies	yes		yes		yes		yes		yes		yes	
Style dummies	yes		yes		yes		yes		yes		yes	
Mean squared error	0.00		0.00		0.00		0.00		0.00		0.00	
Adjusted R^2	0.01		0.01		0.03		0.03		0.01		0.01	
N	160,007		159,164		86,602		86,271		176,203		175,212	

7.4 Table IV: Extended Tests of Strategic Cross-Fund Subsidization

Read:

- Table IV splits the same-family effect according to whether the high-value fund's style is outperforming or underperforming the low-value fund's style.

Table III
Tests of Strategic Cross-Fund Subsidization

This table presents regression results of the tests of cross-fund subsidization. Every month, we partition funds into quartiles with respect to a variable of interest (e.g., Total Fees). A fund is classified as High (Low) if the fund's value for that variable is above (below) the 75th (25th) percentile for the relevant peer group of funds. For Total Fees and Age, the comparison peer group is all funds in the same family. For Year-to-Date Return, the comparison peer group is all funds in the same style (ICDI investment objective code). Note that for the Age criteria, a High value fund is a young fund, while a Low value fund is an old fund. We then construct, for each classification, two sets of High/Low pairs of funds as follows. In the first set, each High fund is matched with all Low funds belonging to the same family. We call these pairs, constructed within the same family, "Actual" pairs. In the second set, each Low fund in every Actual pair is replaced by a matching control fund taken from the remaining sample of funds. The matching control fund is selected randomly from the set of funds of the same style and belonging to the same decile of the variable of interest as the Low fund it replaces (Total Fees, Year-to-Date Return, or Age). We call these pairs "Matched" pairs. For each fund in the pair, we calculate the Net-of-Style Return (defined as the fund's monthly return minus the return for its style) and subsequently the Difference in Net-of-Style Returns, the difference between the Net-of-Style Return of High fund and Low fund i within each pair. Both sets of pairs are added together in the same data set to run our regressions. For each classification, the table shows the results of the following regression (equation (1) in the text):

$$\text{Net.return}_{i,t}^{\text{High}} - \text{Net.return}_{i,t}^{\text{Low}} = \alpha + \beta(\text{Same.family}) + \gamma(\text{Same.style}) + \text{controls} + \epsilon_{i,t,j}.$$

The left-hand side variable is the Difference in Net-of-Style Returns for each pair. Same-Family is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same fund family, and 0 otherwise. Same-style is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same style, and 0 otherwise. Control variables whose coefficients are not shown include, for both the High and the Low funds in each pair, the size of the funds, the age of the funds, the size of the fund's families and the age of the fund's families. t -statistics shown are obtained using heteroskedastic-robust standard errors. The symbols "***", "**", and "*" denote significance at 1%, 5%, and 10% levels, respectively.

	Total Fees		Year-to-Date Return		Age	
	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.
Intercept	0.0012	2.01*	0.0019	2.78*	-0.0007	-0.65
Same family (β)	0.0004	1.75*	0.0006	2.44**	0.0021	5.68***
Same style (γ)	-0.0003	-1.06	-0.0003	-1.06	0.0011	2.40**
Controls	no	yes	no	yes	no	yes
Year dummies	yes	yes	yes	yes	yes	yes
Family dummies	yes	yes	yes	yes	yes	yes
Style dummies	yes	yes	yes	yes	yes	yes
Mean squared error	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R^2	0.01	0.01	0.03	0.03	0.01	0.01
N	160,007	160,164	86,602	86,771	176,003	176,212

Table IV

Extended Tests of Strategic Cross-Fund Subsidization

This table presents regression results of extended tests of cross-fund subsidization. Please refer to Table III for a complete description of how pairs of "High" and "Low" funds are obtained. The table shows the results of the following specification (equation (2) in the text):

$$\text{Net.return}_{i,t}^{\text{High}} - \text{Net.return}_{i,t}^{\text{Low}} = \alpha + \beta_1(\text{Same.family}(\text{ST.RET}_{\text{High}} > \text{ST.RET}_{\text{Low}})) + \beta_2(\text{Same.family}(\text{ST.RET}_{\text{High}} < \text{ST.RET}_{\text{Low}})) + \beta_3(\text{Same.family}(\text{Same.style}) + \gamma(\text{Same.style})) + \text{controls} + \epsilon_{i,t,j}.$$

The left-hand side variable is the Difference in Net-of-Style Returns for each pair. The first variable in the right-hand side (Same-Family (ST.RET_{High} > ST.RET_{Low})) is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same fund family and the style of the High fund is currently performing better than the style of the Low fund. The second variable (Same-Family (ST.RET_{High} < ST.RET_{Low})) takes a value of 1 if the two funds belong to the same fund family and the style of the Low fund is currently performing better than the style of the High fund. (Same-Family (Same-Style)) takes a value of 1 if the two funds belong to the same fund family and the same style. Other variables are defined as before. The table presents also the p -value of the hypothesis test that the two coefficients β_1 and β_2 are equal. The number of observations, N , represents pair-month combinations. t -statistics shown are obtained using heteroskedastic-robust standard errors. Symbols "***", "**", and "*" denote significance at 1%, 5%, and 10% levels, respectively.

	Total Fees		Year-to-Date Return		Age	
	Coeff.	t -Stat.	Coeff.	t -Stat.	Coeff.	t -Stat.
Intercept	0.0009	1.49	0.0016	2.31**	-0.0010	-0.86
Same family (β_1)	-0.0005	-18.67***	-0.0004	-17.30***	-0.0012	-2.53**
Same family (β_2)	0.0004	21.58***	0.0007	21.77***	0.0003	10.70***
Same style & family (β_3)	0.0001	1.16	0.0008	1.83	0.0022	2.85***
Same style (γ)	-0.0004	-0.86	-0.0004	-0.90	0.0010	1.48
Controls	no	yes	no	yes	no	yes
Year dummies	yes	yes	yes	yes	yes	yes
Family dummies	yes	yes	yes	yes	yes	yes
Style dummies	yes	yes	yes	yes	yes	yes
p -value of test that $ \beta_1 < \beta_2 $	0.065*	0.013**	0.001***	0.001***	0.31	0.068*
Mean squared error	0.00	0.00	0.00	0.00	0.00	0.00
Adjusted R^2	0.02	0.02	0.03	0.03	0.04	0.01
N	160,007	160,164	86,602	86,771	176,003	176,212

- For total fees, the coefficients suggest that families favor high-fee funds particularly when low-fee funds' styles are doing better, making performance easier to reallocate.
- For year-to-date return, the same-family effect remains strong and positive in relevant style-state conditions.
- Intrastyle same-family terms are smaller, showing that the pattern is not simply a same-style phenomenon.
- Age results again do not support a clean favoritism interpretation.

Detailed interpretation: The extended tests help distinguish strategic cross-subsidization from ordinary style-level return differences. If families are transferring performance, the effect should depend on the relative performance of the high- and low-value funds' styles. Table IV shows patterns consistent with this intuition for fee and performance classifications.

Economic message: The table strengthens the mechanism story: families seem to reallocate performance when relative style conditions make such reallocation more feasible or valuable.

7.5 Table V: Family Characteristics and Strategic Cross-Fund Subsidization

Read:

- Table V reports the same-family coefficient across different types of fund families.
- Subsidization for high fees and high past performance is more evident in large families and families with many funds.
- Families with above-average size heterogeneity show stronger favoritism, especially for total fees and year-to-date return.
- For age, old families appear to favor young funds in some cases, while young families do not; the direction varies.

Detailed interpretation: Family-level heterogeneity is important because strategic subsidization requires both incentives and opportunities. Large families have more funds across which to transfer performance and more to gain from creating star funds. Heterogeneous families may have clearer high-value and low-value funds, making the tradeoff sharper.

Economic message: The effect is not uniform across the industry. It is strongest where a family has the scale and internal complexity needed to move performance across funds.

Table V
Family Characteristics and Strategic Cross-Fund Subsidization

This table presents regression results of the tests of cross-fund subsidization, broken down across family characteristics. Each cell shows the β coefficient of the specification of Panel A of Table III for different family groups. Please refer to the caption of Table III for a complete description of the variables and how the pairs of "High" and "Low" funds are constructed. Families are divided into groups according to size (top 25 families versus bottom 25), number of funds in the family (more or less than the average of 27 funds), family age (more or less than the average of 13 years), and dispersion of fund size within the family (above and below the average). Dispersion of fund size within the family is calculated as the coefficient of variation of the TNA of funds within the same family. The symbols ***, **, and * denote significance of the β parameter at 1%, 5%, and 10% levels, respectively, using heteroskedastic-robust standard errors.

Panel A				
Families by Size		Families by Number of Funds		
	Top 25	Next 25	Above Average	Below Average
Total fees	0.07**	0.07	0.12***	-0.08**
Year-to-date return	0.29***	0.16*	0.34***	0.09*
Age	-0.06**	0.04	-0.09***	0.09**

Panel B				
Families by Age		Families by Size Heterogeneity		
	Old (> 13 Years)	Young (< 13 Years)	Above Average	Below Average
Total fees	-0.05	0.09***	0.11**	-0.05

Table V: Family Characteristics and Strategic Cross-Fund Subsidization

7.6 Table VI: IPO Allocations across High and Low Funds

Read:

- Table VI studies whether underpriced IPO allocations are directed to high-value funds.
- Panel A describes the IPO sample and reports large average first-day returns, indicating that IPO allocations can be valuable.
- Panel B shows that high-value funds receive more or better IPO allocations under the total-fee and year-to-date return classifications.
- The percentage contribution of IPO underpricing to returns is higher for high-value funds in key classifications.
- IPO allocation results are weakest for the age classification.

Detailed interpretation: Hot IPOs are a scarce resource that can boost fund performance. If families have discretion over which member funds receive IPO allocations, directing hot IPOs to high-value funds is a direct subsidy mechanism. Table VI shows that IPO allocation can explain part of the observed performance gap.

Economic message: This table turns the return-gap result into a mechanism: favoritism can operate through the allocation of valuable investment opportunities, not just through abstract performance shifting.

7.7 Table VII: Opposite Trades and Strategic Cross-Fund Subsidization

Read:

- Table VII tests whether opposite trades among family funds explain part of the same-family return gap.
- Opposite trades occur when high- and low-value funds in the same family trade the same stock in opposite directions.
- The table includes variants based on same-quarter opposite trades and lagged/discrete versions of opposite trades.
- The same-family coefficient is attenuated in some specifications when opposite-trade variables are included.

Table VI
IPO Allocations across High and Low Funds

This table presents an investigation of how IPO allocations differ across "High" and "Low" funds. Please refer to Table III for a complete description of how pairs of High and Low funds are obtained. We obtain data for IPOs from the Securities Data Corporation Platinum database for the 1992 to 2001 period, and calculate the equally weighted average and median first-day return (defined as the percentage price increase from the offer price to the first day closing price). We match these IPOs with the Spectrum Mutual Funds holdings data for the funds in our sample. The table presents, for the different High and Low groups in each classification variable (e.g., fees, performance, age), the time-series and cross-section average of the number of IPOs, as well as the average and median first-day IPO returns. The next-to-last row shows the average total dollar amount of underpricing (the first-day price increase times number of shares held) allocated to each group. The last row shows the contribution of underpricing to fund returns, defined as the average ratio between the dollar amount of underpricing and the fund's previous quarter TNA, for all funds that had positive holdings in any IPO. Note that for the Age criteria, a High value fund is a young fund, while a Low value fund is an old fund. The table also shows the p -value of the hypothesis test that the figures are equal for High and Low funds. The symbols ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel A									
All IPO issues, 1992-2001 (source: SDC)		$N = 5,477$		Value: \$413.2 billion		Average 1 st -day return		14.5%	
IPOs held at quarter-end by funds in the sample (source: Spectrum MF Holdings)		$N = 2,657$		Value: \$26.7 billion		Median 1 st -day return		6.3%	
						Average 1 st -day return		28.0%	
						Median 1 st -day return		12.5%	

Panel B									
N	Total Fees		p -Value Diff	Year-to-Date Return		p -Value Diff	Age		p -Value Diff
	IPOs Held by High Funds	IPOs Held by Low Funds		IPOs Held by High Funds	IPOs Held by Low Funds		IPOs Held by High Funds	IPOs Held by Low Funds	
1,751	1,277	1,666		1,061	2,249		872		
Average 1 st -day return	44.0%	30.6%	< 0.001***	50.5%	37.2%	< 0.001***	43.8%	44.7%	0.72
Median 1 st -day return	20.8%	13.5%	< 0.001***	20.0%	19.4%	< 0.001***	19.2%	17.8%	0.40
Dollar amount of underpricing going to H or L funds (\$ billion)	\$2.22	\$1.96		\$6.39	\$1.04		\$6.56	\$2.30	
Percentage contribution of underpricing to returns of H or L funds (% of TNA)	0.474%	0.091%	0.001***	0.288%	0.169%	< 0.001***	0.322%	0.144%	0.004***

- This suggests that trading coordination may be one channel of cross-fund subsidization.

Detailed interpretation: Opposite trades are important because they may represent either implicit liquidity provision or direct cross-trading. A low-value fund may sell into a high-value fund’s buying pressure, or vice versa. Even when these trades are legal or cost-saving, they can redistribute performance across fund shareholders.

Economic message: The table provides mechanism evidence that family-level trading decisions can shift performance across funds, making internal coordination a potential source of investor-level conflicts.

Table VII
Opposite Trades and Strategic Cross-Fund Subsidization
This table presents regression results of the relationship between common holdings and cross-fund subsidization. Quarterly fund holdings are obtained from Spectrum-Mutual funds database. Please refer to Table III for a complete description of how pairs of “High” and “Low” funds are obtained. For each classification (e.g., fees, performance, and age), we run the following regression (equation (3) in the text):
$$\text{Net.return}_{i,t}^{\text{High}} - \text{Net.return}_{i,t}^{\text{Low}} = \alpha + \beta(\text{Same.family}) + \gamma(\text{Same.style}) + \zeta(\text{Opposite.trades}) + \theta(\text{Opposite.trades} \times \text{Same.family}) + \text{controls} + \varepsilon_{i,t,f,s}$$

The left-hand side variable is the Difference in Net-of-Style Returns for each pair. Same.family is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same fund family, and 0 otherwise. Same.style is a dummy variable that takes a value of 1 if the two funds in the pair belong to the same style, and 0 otherwise. For each classification, we present two different versions of Opposite.trades. Opposite.trades_{sym} denotes the sum, across the two funds High and Low in a given quarter, of the percentage of stocks (in dollar value) for which there were opposite (of symmetric sign) changes in holdings with respect to the total stock holdings (in dollar value) of both funds. Opposite.trades_{asym} denotes the minimum, across the two funds High and Low in a given quarter, of the percentage of stocks (in dollar value) for which there were opposite (of asymmetric sign) changes in holdings with respect to the total stock holdings (in dollar value) of both funds. (Opposite.trades × Same.family) denotes an interaction variable obtained by multiplying the corresponding version of Opposite.trades with the Same.family dummy. Control variables whose coefficients are not shown include, for both the High and the Low funds in each pair, the size of the funds, the age of the funds, the size of the funds’ families, and the age of the funds’ families. Time dummies, family dummies, and style dummies are also included in the regression. To ensure the strategy we study is implementable, the timing of measurement of the variable is as follows. The classification of funds is done at the beginning of each month (we discard the month of January, because its classification is the final classification of the previous year). Opposite.trades is calculated using the corresponding quarter change in holdings. Difference in Net-of-Style Returns is calculated after the end of the month, using compounded 3-month returns. The number of observations, N , represents pair-month combinations. t -statistics shown are obtained using heteroskedastic-robust standard errors. The symbols “*”, “**”, and “***” denote significance at 1%, 5%, and 10% levels, respectively.

	Total Fees		Year-to-Date Return				Age			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Coeff.	t-Stat.	Coeff.	t-Stat.	Coeff.	t-Stat.	Coeff.	t-Stat.	Coeff.	t-Stat.
Intercept	0.0001	0.33	0.0004	0.20	0.0427	16.66***	0.0424	16.50***	0.0154	5.52***
Same.family (β)	0.0001	0.83	0.0001	1.23	0.0058	6.23***	0.0056	6.13***	-0.0003	-0.42
Opposite.trades _{sym} (ζ)	-0.0633	-6.46***	-	-	-0.1890	-8.85***	-	-	-0.0124	-1.46

me	0.0234	1.97**	-	-	0.0459	2.16**	-	-	0.0307	2.93***	-	-
me	-	-	-0.4978	-3.06***	-	-	-1.9707	-5.34***	-	-	-0.0659	-0.42
	-	-	0.3754	1.97**	-	-	0.7637	1.88*	-	-	0.6045	2.72**
	0.0001	0.83	0.0006	0.78	0.0034	3.73***	0.0033	3.60***	0.0011	1.76	0.0011	1.72
	yes		yes		yes		yes		yes		yes	
	yes		yes		yes		yes		yes		yes	
	yes		yes		yes		yes		yes		yes	
	yes		yes		yes		yes		yes		yes	
	0.01		0.01		0.01		0.01		0.01		0.01	
	0.03		0.03		0.09		0.08		0.02		0.02	
	92,117		92,117		58,123		58,123		97,552		97,552	

8 Detailed result map

- **Return evidence.** Same-family high/low pairs show larger performance gaps than matched pairs, especially for high-fee and high year-to-date return funds.
- **Style-state evidence.** Extended regressions show that favoritism depends on relative style conditions, consistent with strategic reallocation.
- **Family-characteristic evidence.** Larger, broader, and more heterogeneous families display stronger subsidization.
- **IPO mechanism.** High-value funds receive more favorable exposure to underpriced IPOs.
- **Trading mechanism.** Opposite trades among funds partly explain same-family performance gaps.
- **Non-result.** The young-fund/old-fund channel is not strongly supported, suggesting that family favoritism focuses on clearly monetizable fee and flow incentives.

9 Short summary

- The paper studies whether mutual fund families strategically favor some funds over others.
- High-value funds are high-fee funds, high recent performers, or young funds.
- The key outcome is the high-minus-low difference in net-of-style returns.

- The key comparison is between actual pairs in the same family and matched pairs from different families.
- High-fee and high-performing funds receive extra performance inside families.
- The effects are about 6–28 basis points per month depending on the classification.
- The age channel is weak and often inconsistent with the hypothesis.
- Large, complex, and heterogeneous fund families show more evidence of favoritism.
- Preferential IPO allocation explains part of the performance reallocation.
- Opposite trades and possible cross-trades also explain part of the performance reallocation.
- The paper contributes to delegated asset management by showing that conflicts arise at the family level, not only at the fund-manager level.

10 Conclusion

10.1 Core conclusion

Gaspar, Massa, and Matos show that mutual fund families appear to favor funds that are more valuable to the management company. High-fee and high-performing funds earn extra net-of-style performance relative to low-value funds in the same family, beyond the gap observed for comparable funds from different families.

10.2 Economic intuition

A mutual fund family earns fees from assets under management. Because investor flows respond strongly and convexly to good performance, especially among already successful funds, a family may gain more by creating a standout winner than by maximizing every fund's return equally. This creates an incentive to shift scarce opportunities and trading advantages toward funds whose performance matters most for family value.

10.3 Incremental contribution recap

The paper's contribution is to identify a family-level agency problem in delegated asset management. It shows that the family form of organization can create incentives to redistribute performance across investors in different funds. This contribution is distinct from fund-level studies of managerial incentives because it treats the fund family as the strategic actor.

10.4 Mechanism and evidence assessment

The return evidence is persuasive because it is supported by two mechanism tests. Underpriced IPOs are allocated more favorably to high-value funds, and opposite trades among family funds help explain return gaps. These mechanisms make the interpretation more concrete: favoritism is not only a statistical return pattern but may arise from observable allocation and trading choices.

10.5 Regulatory implications

The results have direct implications for regulation of cross-trading and internal allocation policies. Cross-trading can reduce transaction costs and benefit fund investors in aggregate, but it can also create opportunities for self-dealing across funds. Regulators must therefore weigh operational efficiencies against conflicts between shareholders of different funds in the same family.

10.6 Investor implications

The results imply that investors should care not only about the fund they buy but also about its position inside the family. A fund's fees, recent performance, and strategic importance to the family may affect the treatment it receives. Investors in low-value funds may bear hidden costs that do not appear as explicit fees.

10.7 Limitations

The paper cannot observe internal communications, allocation committees, or direct managerial intent. Some opposite trades may reflect independent portfolio decisions rather than coordinated favoritism. The results therefore document patterns consistent with strategic cross-fund subsidization rather than proving every individual case of intentional self-dealing.

10.8 Final takeaway

The central lesson is that mutual fund families can operate like internal capital markets for performance opportunities. The family structure can create economies of scale and scope, but it also creates conflicts across fund investors. Understanding delegated asset management therefore requires analyzing both individual funds and the strategic behavior of the fund family that controls them.